



Föreläsning: 7

Rörliga bilder och hur vi visar dom.

Ove Edfors och Viktor Öwall

Institutionen för Elektrovetenskap

Innehåll



- Något kort om rörliga bilder samt lagring och överföring
- Elektronik för bildvisning
- Några nya och mer specialiserade tekniker
- Specialapplikation: Övervakning med intelligent videoprocessing

Motivation



- Moving images/videos are all around
 - TV
 - Mobile phones
 - computer games
- with different standards
 - PAL, Digital TV, DVB-x, HDTV, videos, DVDs,...
- A tremendous development in displays
 - CRT, Plasma, LCD, TFT, DLP, OLED...
 - 3D-displays

Något kort om rörliga bilder samt lagring och överföring



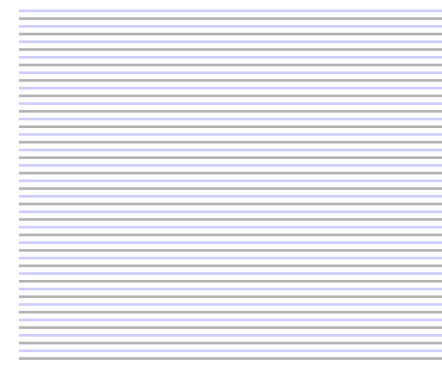
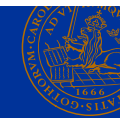
Moving images



A movie is a sequence of images (usually 25 per second or less):



Analog TV-bild (PAL)

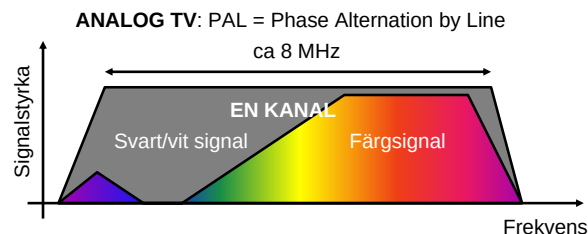


625 linjer, uppdelade på två halvbilder

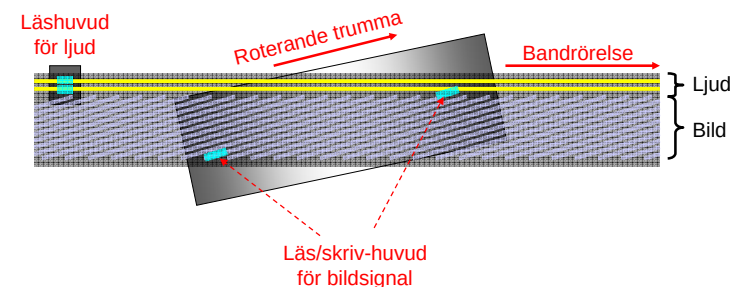
Kallas för INTERLACED

50 halvbilder/sekund = 25 helbilder/sekund

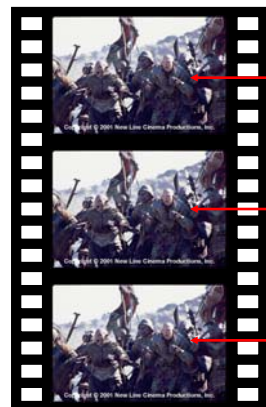
TV-signalens (radio)spektrum



Analog videoinspelning (videoband)



Digital storage of movies



For still images we used the fact that neighbouring pixels are of similar color to do image compression.

We can use that for movies too, but in addition we can use that images following each other in the sequence are quite similar.

ALMOST THE SAME IMAGE ... ONLY MINOR DIFFERENCES. (Not valid during a complete change of camera angle or change of scene, but these are rare.)

Digital storage of movies



Movies (some formats):

MPEG

- **M**oving **P**icture **E**xperts **G**roup

The most famous format for storage of movies. Contains both (moving) image and sound coding. Comes in several different versions, for example:

MPEG-1

- Video CD and MP3

MPEG-2

- Digital Television and DVD.

MPEG-4

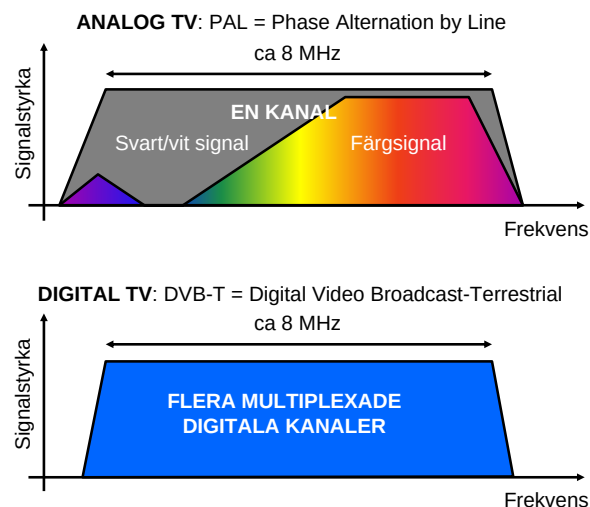
- Fixed and mobile web applications.

QuickTime

- Format developed by Apple Inc.

Used frequently on Macintosh computers and in web or CD-rom based multimedia productions. Many similarities with MPEG-4.

TV-signalens (radio)spektrum



Svenska digitala marknätet (TV)



MULTIPLEX

Flera kanaler på samma frekvensband (DVB-T-signal).





Elektronik för bildvisning (främst rörliga bilder)

2006-10-12

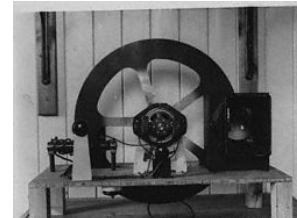
ETI 125 - Föreläsning 7

13

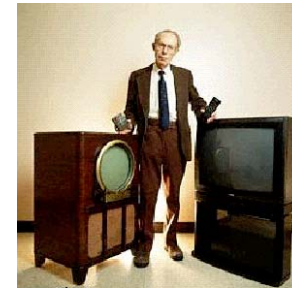
Displays



Mechanical TV system
from 1949



Early B/W TV and
a modern color TV



10 cm thick
50 inch screen

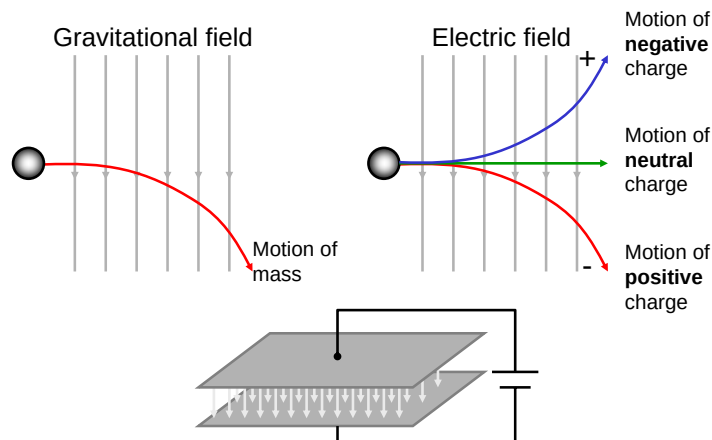


2006-10-12

ETI 125 - Föreläsning 7

14

Compare: Gravity and electric field

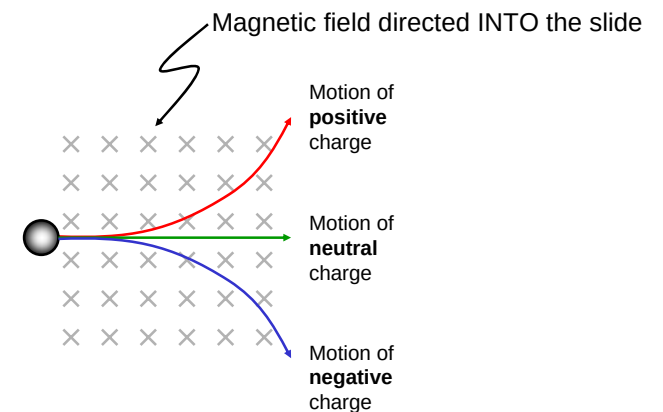


2006-10-12

ETI 125 - Föreläsning 7

15

Magnetic field and forces

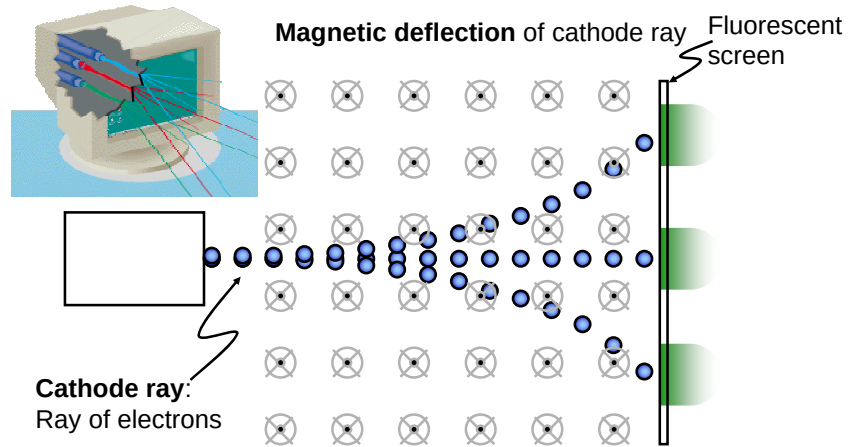


2006-10-12

ETI 125 - Föreläsning 7

16

Application: Cathode ray tube

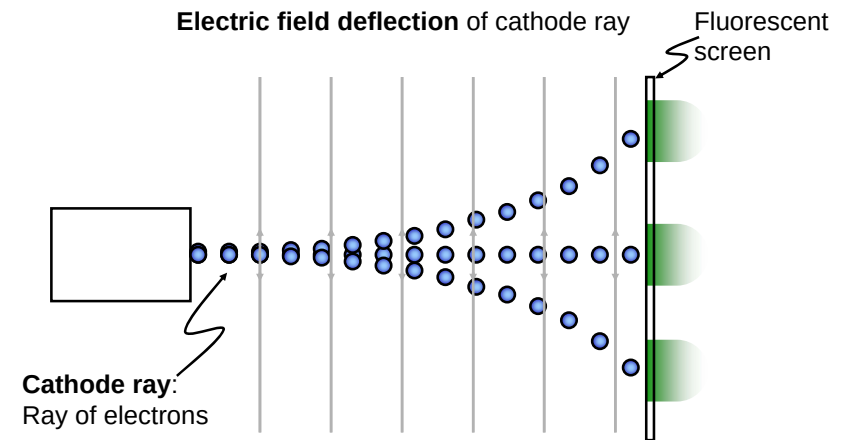


2006-10-12

ETI 125 - Föreläsning 7

17

Application: Cathode ray tube

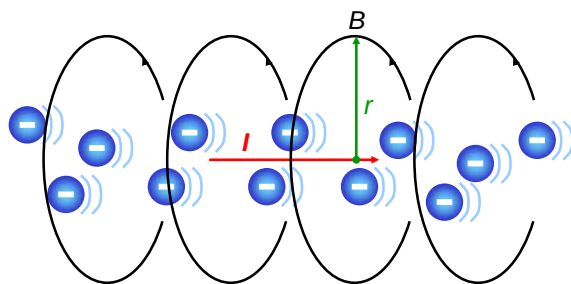


2006-10-12

ETI 125 - Föreläsning 7

18

Magnetic field



Strength of magnetic field at distance r from a "long" straight current:

$$B = \mu_0 \frac{I}{2\pi r} \quad [\text{Tesla (T)}]$$

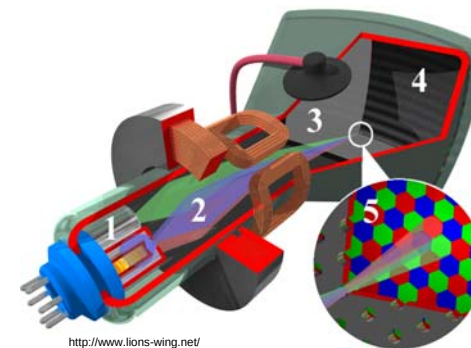
$$\mu_0 = 4\pi \times 10^{-7} \quad [\text{Tm/A}]$$

2006-10-12

ETI 125 - Föreläsning 7

19

Avlänkning i CRT



One set of coils creates a magnetic field that moves the electron beam vertically, while another set moves the beam horizontally. By controlling the voltages in the coils, you can position the electron beam at any point on the screen.

2006-10-12

ETI 125 - Föreläsning 7

20

Traditional Monitors - CRT



CRT - Cathode Ray Tubes



Flat screens



- LCD –based
- Plasma
- Projection based – flat?
- SED
- OLED

Displays



- LED displays - 7 segment

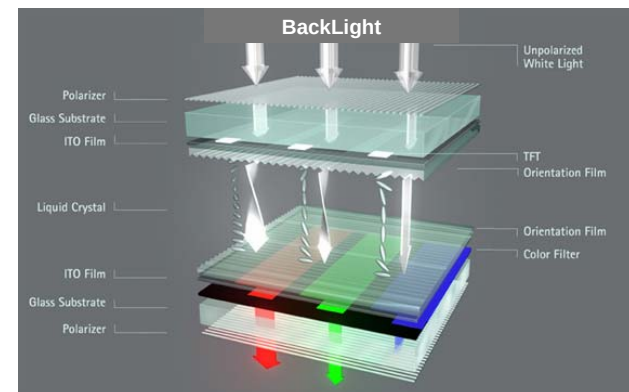


- **LCD (liquid crystal display)**

- liquid crystals are changed by electric current
- liquid crystal materials emit no light of their own
 - Backlit vs. Reflective



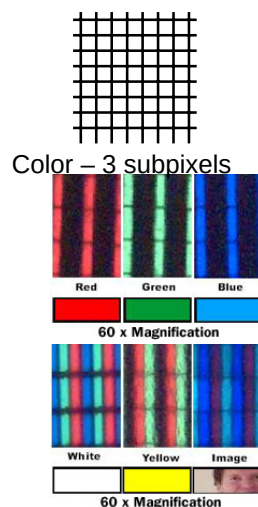
LCD displays - principle



- The light is polarized
- The LCD changes polarization of the light depending on the voltage applied
- The light is passing or not depending on the polarization

LCD displays

- Signal down the correct column and the correct row, intersect at the designated pixel
- Passive matrix**
 - slow response time and imprecise voltage control
- Active Matrix**
 - thin film transistors (TFT), active component
 - 256 levels of brightness per pixel
 - color: $256 \times 256 \times 256 = 16.8$ million colors



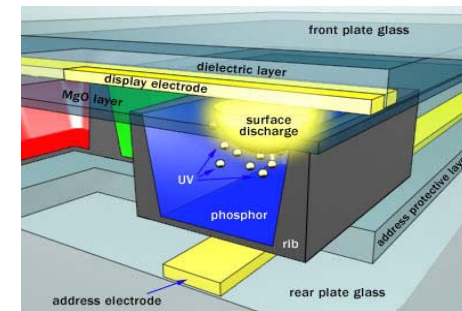
Plasma cells

plasma, a gas made up of free-flowing **ions** (electrically charged atoms) and **electrons**.

A plasma cell is filled with a mix of inert gases, e.g. neon or xenon. Passing electricity through the gas causes it to become temporarily charged, plasma. Light is produced when the gas spontaneously discharges.

Those discharges results in that UV light is emitted

The cell is coated with phosphors material which turn UV into visible light, compare fluorescent tube.



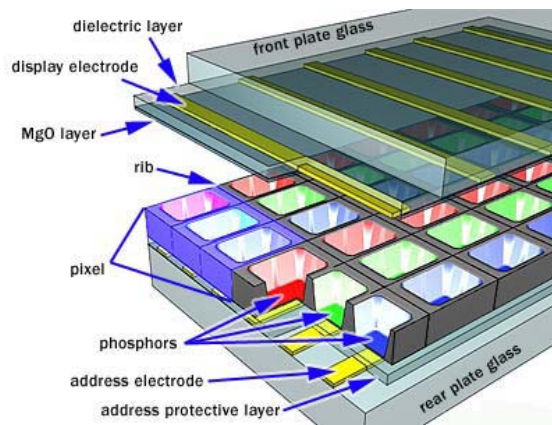
Plasma displays

To ionize the gas in a particular cell, the plasma display's computer charges the electrodes that intersect at that cell.

By varying the pulses of current flowing through the different cells, the intensity is controlled.

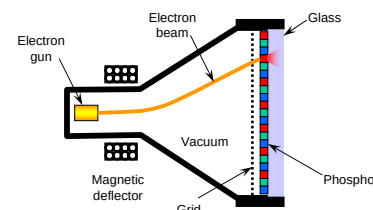
+thin
+viewing angle
+colour performance

-price
-burn in
-lifetime

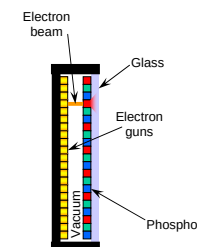


Surface-Conduction Electron-Emitter Display

Standard Cathode Ray Tube (CRT)

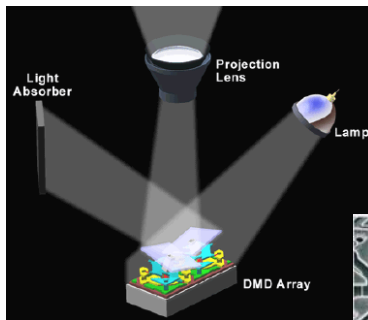


Surface-Conduction Electron-Emitter Display (SED)



First products came out in 2005
Fast update (like CRT)
Low energy consumption

DLP television –digital light processing



The leg of an ant rests on this DMD mirror array, courtesy of TI



In a digital micromirror device (DMD) each pixel, **up to 2 million**, is represented by a mirror attached to a yoke and a hinge. The mirrors are moved to on and off positions.

- Lower price for larger displays
- No burn outs – one lamp can be changed

DLP applications



56-inch DLP/Back projection TV
50cm deep

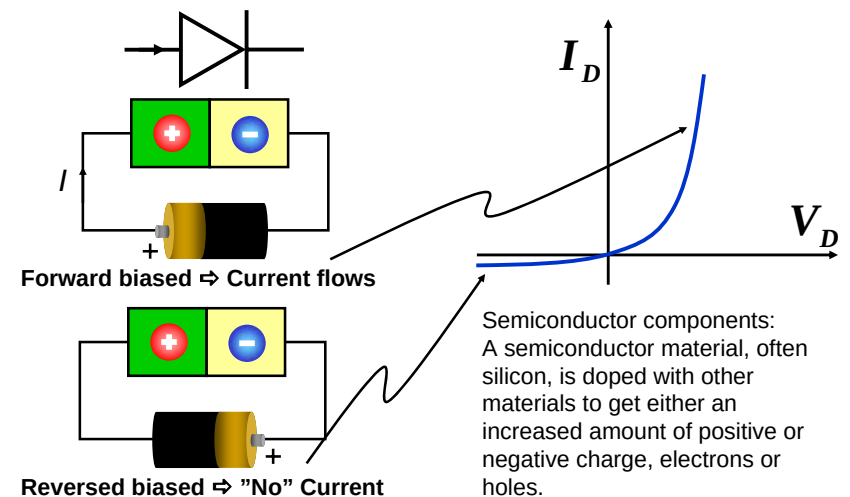


DLP Projector

Några nya och mer
specialiserade tekniker



Semiconductor components: Diode



LED – Light Emmitting Diode

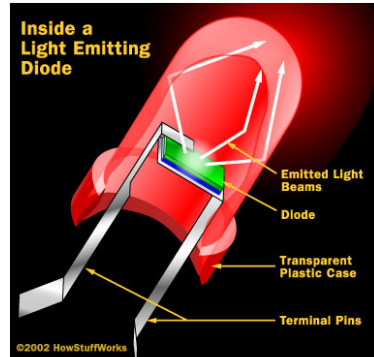


All normal diodes emit light, mostly infrared

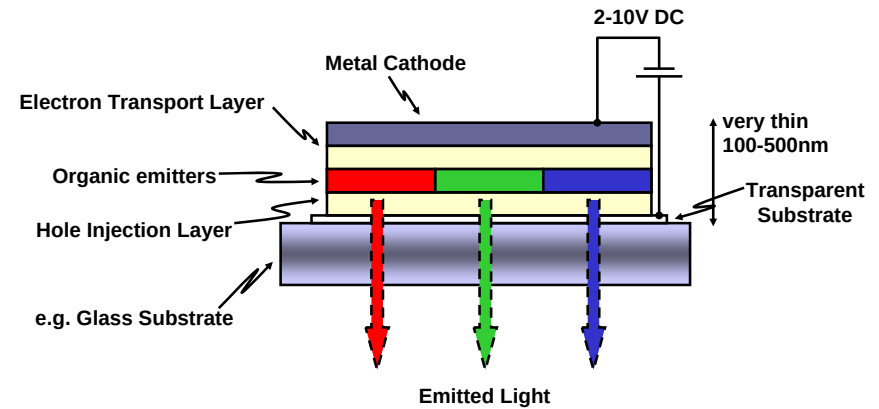
LEDs are specially constructed to release a large number of photons outward as visible light.

Advantages

- no filament that will burn out
- small plastic bulb
- efficiency, little heat



Organic LED

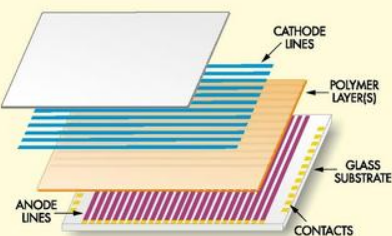


Kodak scientists Ching Tang and Steve Van Slyke demonstrated an efficient, low-voltage OLED in 1987.

OLED displays



Passive OLED display, a pixel is turned on when respective cathode and anode lines are turned on



Active displays also includes a TFT layer that controls each pixel.

OLED properties & usage



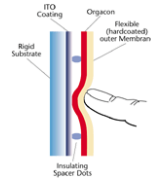
- No backlight needed
- large view angle
- low power
- thin
- fast
- flexible
- cheap?



James Bond
Die Another Day

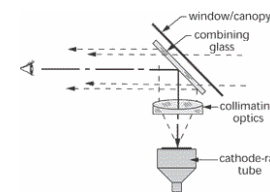
Touch screen

- Resistive
 - press layers together change, detect electrical field
 - reduced clarity and steal light, ca 70% through
 - cheapest
- Capacitive
 - charge transferred to "finger"
 - reduced clarity and steal light, ca 90% through
- Surface acoustic wave
 - acoustics waves transmitted and changes detected
 - usually more expensive



Head-up display

The principle



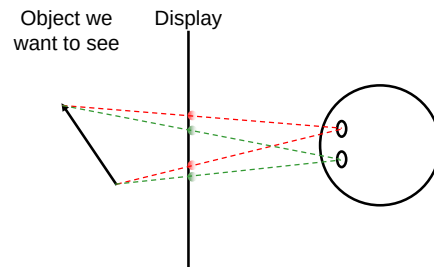
Helmet-mounted display



Chevrolet Corvette



3D displays – basic principle

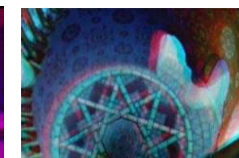


The left and right eye must see different 2D representations of the object

3D displays

"Simple version" – use color filters on left and right eye!

Single image consisting of "two images" ... one red one blue/green



"Cheap" but poor color representation.



3D displays



Alternating left/right view.

The display alternates between the image for the left and the right eye. Synchronized special glasses lets each eye see only its image.

Better quality than color filters, and still "relatively" cheap.



2006-10-12

ETI 125 - Föreläsning 7

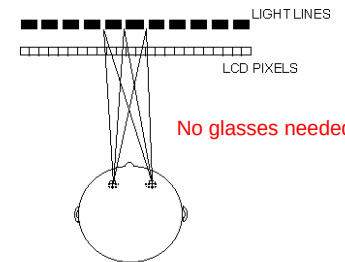
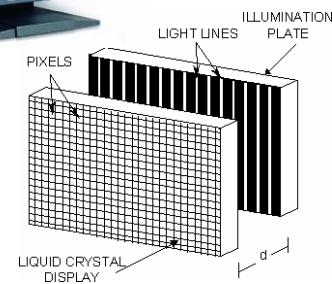
41

3D displays



Can this be true?

Parallax illumination



L/R-Eyes See Every-Other Column

2006-10-12

ETI 125 - Föreläsning 7

42

Low cost paper electronics



Display printed on paper.
Acreo, Norrköping.

2006-10-12

ETI 125 - Föreläsning 7

43

Electronic ink



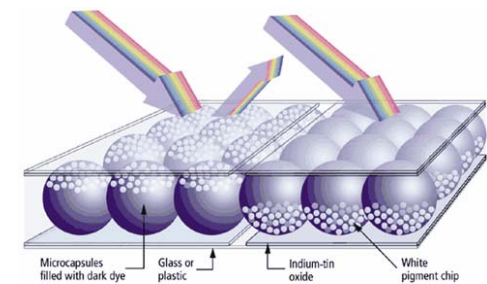
Paper that "prints itself"



Electronic ink display.
E Ink Corp. and
Philips Electronics



Flexible Displays



2006-10-12

ETI 125 - Föreläsning 7

44

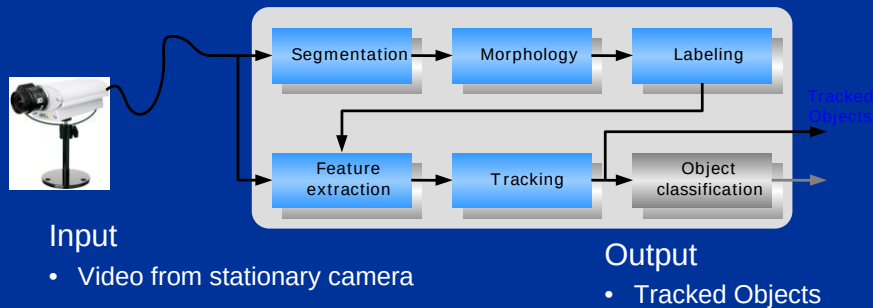
Endoscopy, looking inside the body

- swallow camera in glas cylinder
- 2 images per second
- moving mirrors for circular scan
- 24 hours
- Wireless transmission to belt



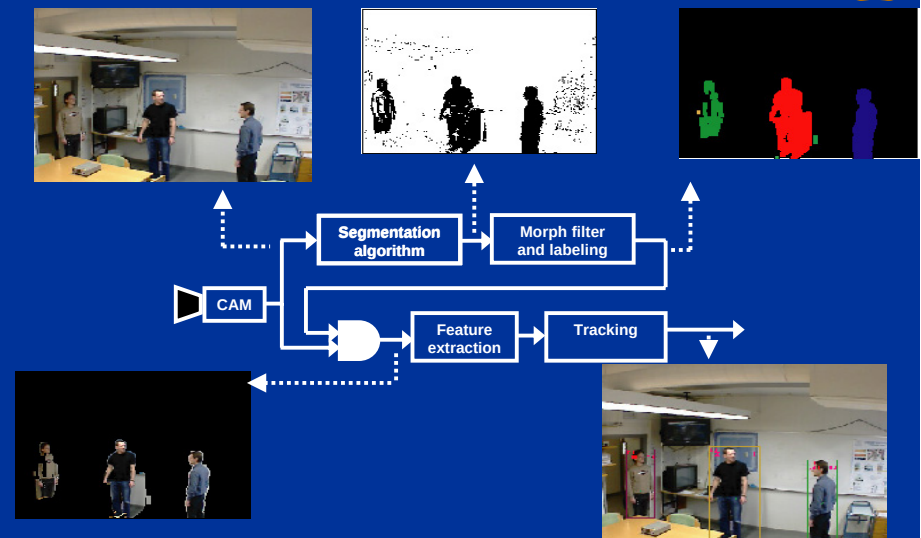
Specialapplikation: Övervakning med intelligent videoprocessning

Nowhere to Hide – an Intelligent Surveillance System

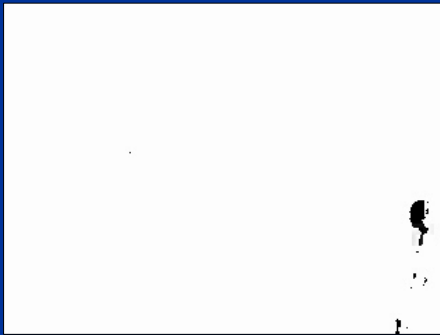


Growth of surveillance systems

Nowhere to hide – An Intelligent Surveillance System



Segmentation – distinguish between foreground and background



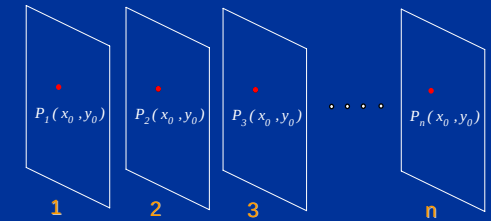
Two problems:

- pixel values change over time
- pixel values may change because of "several backgrounds"

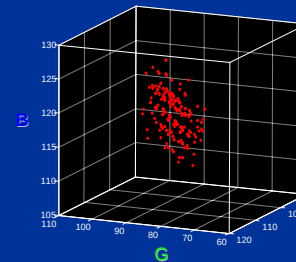
Background Modeling



Sample background environment in the digital lab

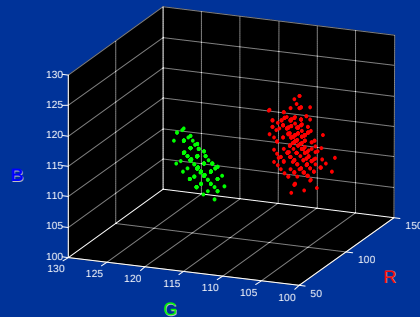


Consecutive Video Frames



Pixel values taken from same location in consecutive video frames looks like a Gaussian distribution in RGB color space

Multi-modal Background



Background pixels such as lake surface and swaying trees have the property of two distributions requiring two Gaussian to model

Tracking/Classification



Nästa föreläsning



- Digitalkamera och bildlagring
 - Bildsensorer
 - Bildkodning
 - Minnestekniker